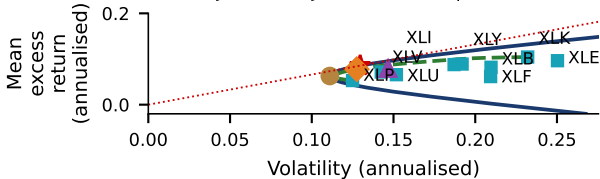


Nine sector ETFs, monthly excess returns Jan 1999 - Jul 2026 (in-sample)



- Frontier, short sales allowed
- Frontier, long-only
- GMV: Sharpe 0.57

- Tangency (MV): Sharpe 0.66
- Tangency, long-only: Sharpe 0.61
- 1/N: Sharpe 0.54